Ashutosh Singh Rawat
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EDUCATION

Vellore Institute of Technology Aug 2022 – May 2026
Bachelor of Technology
Major in Computer Science and Engineering
Cumulative GPA: 8.86

TECHNOLOGIES AND TOOLS

- **Languages:** Python, Java, SQL.
- **Tools/Cloud:** AWS Cloud, Docker.
- **Back-end & APIs:** Fast API, Django, Node.js, Express.js, gRPC, WebSockets, REST APIs.
- **Databases:** Databricks, MongoDB, Relational Databases (MySQL, PostgreSQL).
- **Data Engineering**: Apache Spark (PySpark), Apache Kafka, ETL Pipelines, Databricks on AWS.
- **Machine Learning & Statistics:** Scikit-learn, TensorFlow, PyTorch, Feature Engineering, Computer Vision, SHAP, Grad-CAM.
- **Generative AI**: OpenAI APIs, LangChain, RAG, Prompt Engineering, AI Agents, GitHub Copilot
- **Relevant Coursework:** Data Structure and Analysis of Algorithm, Operating Systems, Computer Networks, Database Management, Real-time Processing, Explainable AI, Model Deployment.

CERTIFICATIONS

- **Cloud computing**, NPTEL
- **AWS Academy Cloud Foundation**, AWS
- **AWS Certified Cloud Practitioner,** AWS

WORK EXPERIENCE

AI System Evaluation Intern Jul 2026 – Present
AirDawg Labs – Caudal AI

- Engineered datasets and edge-case scenarios, identifying LLM failure points across tasks and models to get reduction in critical errors.
- Created prompt engineering strategies and AI-assisted evaluation workflows to improve LLM performance, reliability, and response quality.
- Evaluated and executed structured test scenarios for AI models, validating functionality, edge cases, and system reliability.

AI Engineer Intern Apr 2026 – Present
Indian Institute of Technology Delhi (IIT Delhi)

- Optimized predictive ML models across 3 stages: preprocessing, feature engineering, evaluation for highly accurate data analytics
- Built RAG pipelines with LLMs, Lang Chain, and Vector DBs for semantic search to boost information retrieval speed.
- Programmed Python AI workflows for scalable data preparation, training, and deployment ensuring smooth production level transitions.

AI/ML Intern Nov 2024 – Feb 2025
Huwats India Pvt. Ltd – Bengaluru (Remote)

- Pioneered a real-time Python/OpenCV object detection interface with >90% accuracy for precise automated spatial analysis.
- Applied contour detection, image segmentation, and adaptive thresholding to improve edge precision by 25% and enhance spatial analysis.
- Deployed image processing pipelines to improve processing efficiency and ensure consistent real-time object detection performance.

PROJECTS

Omni Serve - ENTERPRISE AI API GATEWAY GitHub Repo
Event-Driven Machine Learning (ML) Microservice Architecture

- Architected a centralized AI API gateway routing 50+ ML requests/sec via Fast API and gRPC to eliminate bottlenecks and reduce latency.
- Orchestrated Docker/K8s microservices and CI/CD pipelines, cutting deploy time 50% to streamline automated continuous delivery.
- Designed a Kafka telemetry pipeline ingesting 10GB+ daily data with Redis rate limiting for reliable real-time system monitoring.
- **Tech Stack**: Python (Fast API), gRPC, Apache Kafka, Redis, PostgreSQL, Grafana, Docker, Kubernetes (K8s).

Med AI - UNIVERSAL DISEASE PREDICTOR GitHub Repo
AI-Powered Medical Imaging Diagnostic System

- Developed a full-stack Fast API/MERN diagnostic platform integrating models (ResNet50, VGG16) for multimodal medical imaging.
- Applied Grad-CAM Explainable AI to visualize model predictions, significantly improving diagnostic transparency.
- Crafted OpenCV computer vision pipelines to enhance image quality and optimize machine learning feature extraction.
- **Tech Stack**: Python, TensorFlow, PyTorch, OpenCV, Fast API, React, MongoDB.

CO-CURRICULARS

- Co-founder, Pahadi Club: Scaled a cultural club, organizing 5+ large-scale events for 300+ participants that significantly boosted cross-cultural engagement and student participation by 40%.
- National Service Scheme (NSS): Planted 10,000+ trees and raised awareness for 500+ students during major environmental sustainability drives